Remainders, Powers, and Operations

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What Can a Square Leave Behind?

(a) Which remainders can a square leave on division by 8?

(b) Prove that no integer of the form $8k + 7$ is a sum of three integer squares.

Multiplication Rearranges the List

(a) Find the multiplicative inverse of each of $1, 2, 3, 4, 5, 6$ modulo 7.

(b) If 7 does not divide a, show that $a, 2a, \ldots, 6a$ have distinct nonzero remainders modulo 7. Deduce that $a^6 \equiv 1 \pmod 7$.

Long Sums, Short Products

For a positive integer m, factor $t^m - 1$ using a geometric sum. When m is odd, also factor $t^m + 1$. Use your identities to exhibit nontrivial factors of $1 + 10^2 + 10^4 + 10^6 + 10^8$.

Primes Made from Factorials

Express 5 and 7 as quotients of products of factorials of primes. Only $2!, 3!, 5!, 7!$ may be used, and repetitions are allowed. Can you express $5/7$ in the same way?

An Unfamiliar Addition

For integers a, b, define $a \star b = a + b + 1$. Is $\star$ associative? Does it have an identity element? Does every integer have an inverse? Find a bijection $\phi : \mathbb{Z} \to \mathbb{Z}$ such that $\phi(a \star b) = \phi(a) + \phi(b)$.

Alternating Digits

How many primes have decimal expansions consisting of alternating 1's and 0's, beginning and ending with 1?

Source: Putnam 1989 A1

A Fourth Power and a Power of Four

Determine all positive integers n for which $n^4 + 4^n$ is prime.

All Rationals from Prime Factorials

Show that every positive rational number is a quotient of products of factorials of primes, with repetitions permitted. For example, $10/9 = 2!\,5!/(3!\,3!\,3!)$.

Source: Putnam 2009 B1